

COMP3052.SEC Computer Security

Session 15-2 Crypto II: EXTRA MATHS MATERIALS

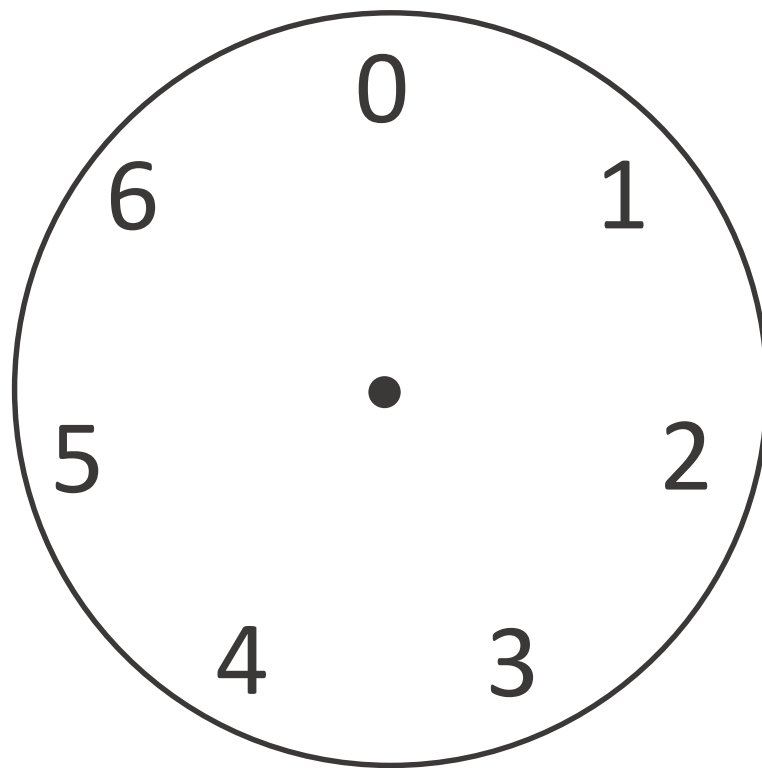


Acknowledgements

- Some of the materials we use this semester may come directly from previous teachers of this module, and other sources ...
- Thank you to (amongst others):
 - Michel Valstar, Milena Radenkovic, Michael Pound, Dave Towey ...

Modular Arithmetic 模运算

- A system of arithmetic based around cycles of numbers 一种基于数字循环的算术体系
- Numbers modulo n are a **finite field** 数字模 n 是一个有限域



The set of numbers
modulo 7

The Congruence Relation

$$a \equiv b \pmod{n}$$

$$a \pmod{n} = b \pmod{n}$$

余数相同:

举例:

$15 \pmod{7} = 1 \pmod{7} \rightarrow$

$$15 \equiv 1 \pmod{7}$$

Equivalences

$$((a \bmod n) + (b \bmod n)) \bmod n = (a + b) \bmod n$$

$$((a \bmod n) \cdot (b \bmod n)) \bmod n = (a \cdot b) \bmod n$$

Multiplication Example

Rule:

$$((a \bmod n) \cdot (b \bmod n)) \bmod n = (a \cdot b) \bmod n$$

Example: $(29013 \cdot 1123) \bmod 7$
 $32581599 \bmod 7$

Multiplication Example

Rule:

$$((a \bmod n) \cdot (b \bmod n)) \bmod n = (a \cdot b) \bmod n$$

Example: $(29013 \cdot 1123) \bmod 7$

$$32581599 \bmod 7$$

Or: $((29013 \bmod 7) \cdot (1123 \bmod 7)) \bmod 7$

$$(5 \cdot 3) \bmod 7$$

$$15 \bmod 7 = \mathbf{1}$$

Exponentiation

Example: $13^{11} \bmod 7 = ?$

Exponentiation

Example: $13^{11} \bmod 7 = ?$

$$((13^2 \bmod 7) \cdot (13^9 \bmod 7)) \bmod 7$$



$$169 \bmod 7 = 1$$

Exponentiation

Example: $13^{11} \bmod 7 = ?$

$$\left((13^2 \bmod 7) \cdot (13^9 \bmod 7) \right) \bmod 7$$

$$\left(1 \cdot (13^9 \bmod 7) \right) \bmod 7$$

$$\left(1 \cdot (13^2 \bmod 7) \cdot (13^7 \bmod 7) \right) \bmod 7$$

etc.

Exponentiation

Example: $13^{11} \bmod 7 = ?$

$$\left((13^2 \bmod 7) \cdot (13^9 \bmod 7) \right) \bmod 7$$

$$\left(1 \cdot (13^9 \bmod 7) \right) \bmod 7$$

$$\left(1 \cdot (13^2 \bmod 7) \cdot (13^7 \bmod 7) \right) \bmod 7$$

etc.

$$\left(1 \cdot (13^1 \bmod 7) \right) \bmod 7$$

$$13 \bmod 7 = \mathbf{6}$$

Logarithms

- A logarithm is the inverse function to exponentiation: 对数是指数运算的逆运算：

$$a^b = c$$

$$b = \log_a(c)$$

- This is easy to compute even for large numbers

Discrete Logarithms

- When operating mod n , we call the operation a **discrete logarithm**: 离散对数运算

$$a^b = c \pmod{n}$$

$$b = \text{dlog}_{a,n}(c)$$

Example:

$$7^2 = 4 \pmod{9}$$

$$2 = \text{dlog}_{7,9}(4)$$

Discrete Logarithms

- Discrete logs are much harder to compute

$$3^? = 1 \pmod{7}$$

$$? = \text{dlog}_{3,7}(1)$$

Discrete Logarithms

- Discrete logs are much harder to compute

$$3^? = 1 \pmod{7}$$

3的几次方 mod 7 等于 1 ?

$$? = \text{dlog}_{3,7}(1)$$

Brute force:
只能暴力破解

$$3^1 = 3 \pmod{7}$$

$$3^2 = 2 \pmod{7}$$

$$3^3 = 6 \pmod{7}$$

$$3^4 = 4 \pmod{7}$$

$$3^5 = 5 \pmod{7}$$

$$3^6 = 1 \pmod{7}$$

Summary



- This very brief session was really just to give you a taste (hopefully a reminder) of the kind of maths to be ready to do